

Week 4 - part 5

Quantum Computing

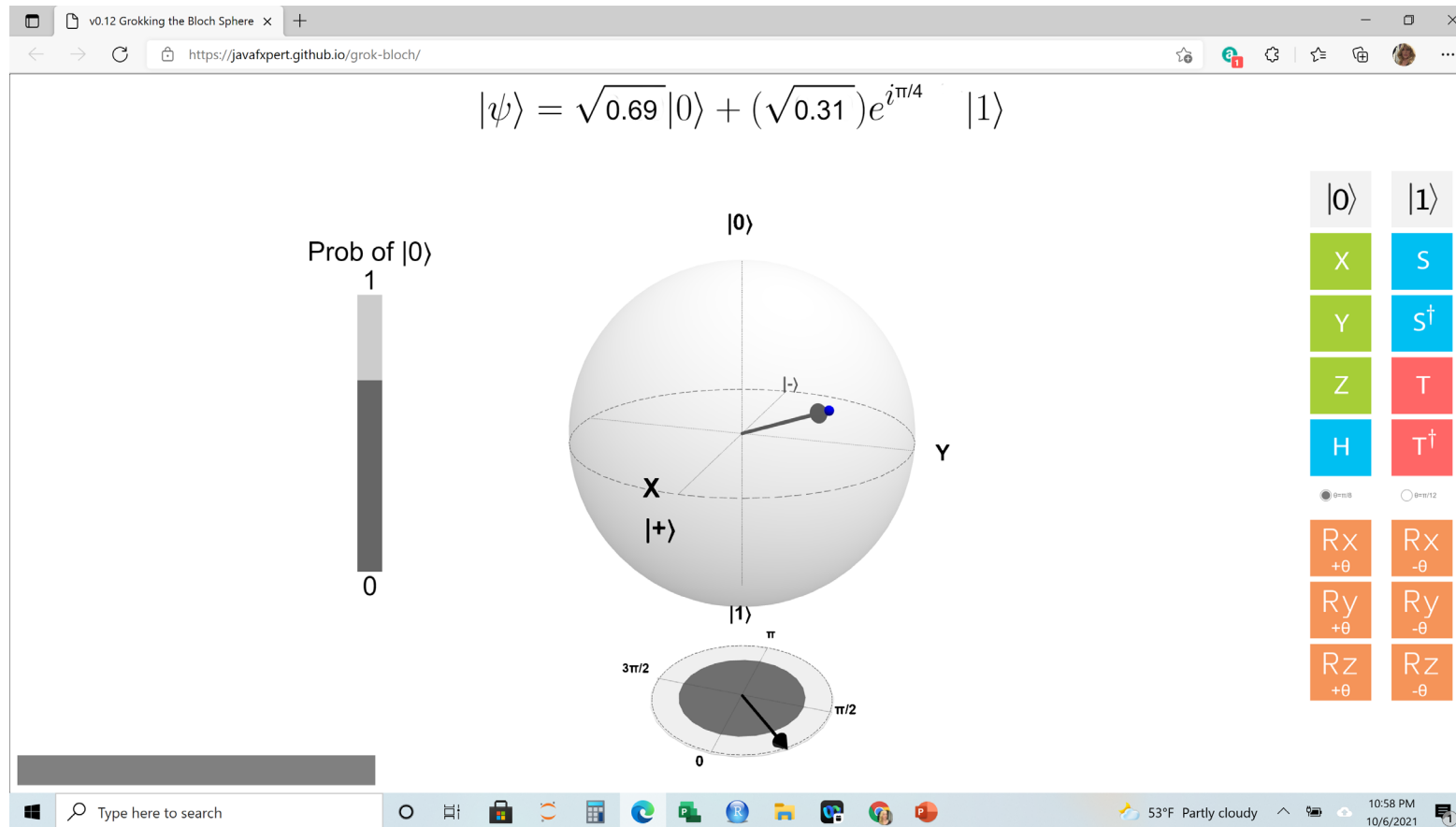
Quantum Logic and Gates

Visualizing Gate Effects, QKD, and Getting your IBM Quantum Account

Nella Ludlow, PhD
Wright State University

Gronk the

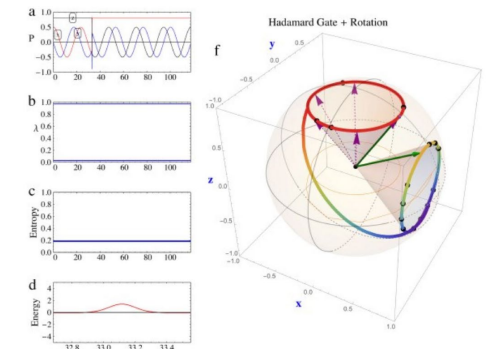
<https://javafxpert.github.io/grok-bloch/>



Professor Nella Ludlow - Wright State University

The Hadamard gate

While the Pauli gates are a lot like classic logic gates in some respects, the Hadamard gate, or *H-gate*, is a bona fide quantum beast. It shows up everywhere in quantum computing, and for good reason. The Hadamard gate has the characteristically *quantum* capacity to transform a definite quantum state, such as spin-up, into a murky one, such as a superposition of *both* spin-up and spin-down at the same time.



QKD - Quantum Key Distribution



<https://youtu.be/Vj4Hk1W4VQ0>

IBM Quantum Composer

- <https://quantum-computing.ibm.com/composer>

